CancerOS Database and API Contracts Specification

Document ID: COS-SPEC-008

Version: 1.0

Date: 22 October 2026

Status: Proposed implementation baseline

Scope: Breast cancer research and hackathon prototype

1 Purpose and contract authority

CancerOS shall use PostgreSQL for typed research records and immutable snapshots, private object storage for image and geometry bytes, and FastAPI for authorized REST operations. This specification defines persistence, requests, responses, asynchronous work, and updates shared by the frontend, scientific workers, simulator, and interpretation engine. It specifies intended behavior; it does not certify an existing implementation.

The first implementation shall support a verified MRI baseline, imported masks, physical measurements, five hypothetical scenario families, dated clinical observations, state navigation, evidence-linked interpretations, and an independently verified real longitudinal case. Dataset size targets are capacity assumptions. Neither clinical validity nor source completeness follows from accepting a record into a database.

MUST and SHALL indicate release requirements. SHOULD indicates a default that may be changed through a recorded design decision. All sample values and identifiers in this document are illustrative contract fixtures, not patient evidence.

Reviewed inputs are COS-PRD-001 and COS-SPEC-003 through COS-SPEC-007, all dated 22 October 2026. Their proposed requirements are reconciled below. Technical standards inform enforcement mechanisms [R1–R4]; application-specific limits and workflows are CancerOS design decisions.

2 Review decisions and corrections

Issue in the source specifications	Binding contract decision
Short and long provenance names coexist	Persist REAL_OBSERVED, REAL_DERIVED, SYNTHETIC_GENERATED, INTERPOLATED, MODEL_PREDICTED. Map to short UI labels only at presentation.
A state can contain measured and derived material	State context is OBSERVED or SYNTHETIC in P0; every member retains its own provenance. A derived mesh does not make its source MRI synthetic.
Visits and states appear interchangeable	Visit is a source occasion; state is an immutable assembled snapshot at a selected time and data revision. Each observation retains its own time.
Synthetic visits are allowed in the dataset document	Store synthetic occasions only inside a branch. No branch-owned visit or observation enters the observed timeline.
Baseline is counted inside each four-state branch	A branch references one shared published real baseline and has at least three generated follow-up states. This is four timeline positions, not four independently real visits.
Time is sometimes described as a required timestamp	Allow exact timestamp, date, source-relative day, or unknown time. Never invent midnight, a timezone, or a calendar date.
Clinical corrections and historical immutability both required	Append a revision or retraction. Old snapshots preserve original member IDs. New snapshots incorporate accepted corrections.
Modality status mixes availability and origin	Separate availability from provenance; SYNTHETIC is an origin, not an availability value.

Issue in the source specifications	Binding contract decision
A source visit is required for every observed state	Permit lab-only and other dated observed states without a clinical visit. Provenance and a valid time reference are still required.
Framework and provider candidates sound like installed capabilities	Register only verified model versions. P0 does not require a segmentation checkpoint, prediction service, or named LLM entitlement.
Quality validation could be confused with clinical validation	QC PASS means the named engineering checks passed. It is not a statement of clinical accuracy.

Anatomical lesion contours, functional analysis masks, and bounding boxes are distinct annotation classes. A checksum proves byte integrity, not patient identity, annotation meaning, license permission, or scientific correctness. Dataset rights and actual case linkage remain import gates; this contract does not reassert source collection counts or licensing claims as established facts.

3 Identity and ownership model

Use one research project for the hackathon, with explicit project ownership on patient-bearing tables. This is a small access boundary, not a full multitenant hospital platform. Internal IDs are UUIDs generated by the server. Display labels are pseudonymous and have no identifier semantics.

Every patient-owned row carries project_id and patient_id. Every branch-owned row additionally carries branch_id. Define UNIQUE(project_id,id) on project resources and UNIQUE(project_id,patient_id,id) on patient resources. Use composite foreign keys so a lesion, study, state, artifact, and observation cannot be linked across patients or projects. Branch membership requires FOREIGN KEY(project_id,patient_id,branch_id) to a matching branch. Ownership columns are immutable.

Branches may reference their own baseline from the observed context through explicit baseline links. Ordinary branch member links require exact ownership equality. Arbitrary cross-context references are forbidden. No browser request may choose project ownership through a trusted request-body field; the authorized route and session determine it.

Pseudonymous source mappings are separately protected. Identity merges require administrator-reviewed source evidence, a recorded equivalence assertion, and reconciliation of evaluation groups. Similar-looking demographics do not establish identity. Uncertain duplicates remain quarantined from evaluation claims.

4 Common database types and rules

Type or field	Definition
id and foreign IDs	uuid; not null except explicitly optional relations
recorded_at	timestamptz not null; server transaction time
revision	bigint positive; optimistic concurrency counter on mutable aggregates
code	text from a versioned controlled dictionary; no PostgreSQL enum migrations required
quantity	numeric(24,9), finite, with dictionary-defined bounds; JSON decimal string
hash	lowercase hex text matching 64 characters for SHA-256
metadata or config	jsonb; schema version required, bounded depth and size; never an untyped replacement for clinical columns
record lifecycle	draft, accepted, rejected; corrections and retractions are separate append-only records
quality	pending, pass, warning, fail, not_applicable; check details and reviewer retained
publication	draft, validating, ready, published, rejected, withdrawn

Tables are in schemas access, source, clinical, imaging, twin, intelligence, and ops. Tables listed below have id and recorded_at unless a composite key is stated. A question mark denotes nullable. Columns not marked nullable are required. Foreign-key names refer to the listed canonical table even when schema qualification is omitted for readability.

No hard deletion through the public API. Restricted rows remain auditable; lawful removal is an administrator maintenance workflow with retention review, artifact deletion/tombstones, backup handling, and a retained non-sensitive audit record. Withdrawn resources disappear from default viewer queries but are not silently relabeled missing.

4.1 Time references

Use a shared time_references table: id, patient_id, branch_id?, basis, timestamp_value?, date_value?, relative_day?, anchor_time_id?, precision, time_domain, source_locator?, recorded_at. time_domain is source_calendar, source_shifted, source_relative, or simulation. basis is timestamp, date, relative_day, or unknown. Exactly the matching value column is populated; unknown has no value. Relative days are numeric(12,4) and require an explicit anchor. Timestamp precision records whether the source was minute- or second-resolved. Date values are not converted into invented timestamps. Unknown dates cannot become ordered state anchors.

Cross-row triggers reject cycles in time anchors and incompatible patient/branch anchors. The observed baseline anchor is the only allowed observed anchor for branch-relative simulation times. Compare only within compatible time domains or an explicit verified conversion. Exact timestamps normalize to UTC while retaining source offset metadata where known. De-identification shifts are not represented as actual calendar dates. Observation time and recorded_at serve different purposes.

4.2 Clinical coding and missingness

concept_definitions(code_system,code,version,label,kind,canonical_unit?,value_type,lower_bound?,upper_bound?,conversion_rules_version?) has a composite unique key on coding identity. Use a local code namespace where standardized mappings have not been verified. Store original value, original unit, and raw source locator before conversion. A numeric value is never silently replaced by zero.

missing_reason is unknown, not_recorded, not_collected, not_applicable, below_detection, or withheld. Numeric observations require exactly one of a quantity, a censored value with comparator and threshold, a coded/text value permitted by their dictionary, or a missing reason. A below-detection result retains its comparator and threshold; it is not an exact concentration. Reference intervals include their source, population/context, unit, and version. Out-of-range is different from invalid input.

5 Access and source schemas

Table	Required columns and optional fields	Constraints and purpose
access.projects	name, status	Project boundary
access.users	issuer, subject, status	Unique issuer and subject; external identity provider owns credentials
access.memberships	project_id, user_id, role	Unique project and user; role viewer, editor, admin
access.patient_grants	project_id, patient_id, user_id, permission	Optional narrower access; default deny without membership and matching policy
source.licenses	identifier, terms_url?, terms_hash?, review_status, reviewed_by?, reviewed_at?, permitted_uses jsonb	reviewed status requires reviewer, timestamp and terms evidence
source.dataset_sources	project_id, name, source_url, release_label, license_id, metadata_hash	Unique project, source URL and release label
source.research_patients	project_id, display_code, identity_group_id, access_status, data_revision	Unique display code within project; no direct identifiers
source.identity_groups	project_id, split, split_version, identity_review_status	split train, validation, test, demo, unassigned; descendants inherit
source.source_patient_mappings	project_id, patient_id, dataset_source_id, source_key_ciphertext, source_key_hmac, key_version, linkage_status	Unique project, dataset and HMAC; never expose ciphertext or HMAC to viewers

Table	Required columns and optional fields	Constraints and purpose
source.identity_assertions	project_id, identity_group_id, mapping_id, evidence_artifact_id, reviewer_id, decision	Append-only reviewed equivalence evidence
source.import_batches	project_id, manifest_artifact_id, status, submitted_by, request_hash, validation_report_id?	Manifest and processing results retained
source.source_locators	project_id, artifact_id, locator_type, locator jsonb, record_hash?	row/sheet, JSON Pointer, DICOM instance, or text span; versioned locator schema
source.provenance_records	project_id, kind, source_locator_id?, processing_run_id?, simulation_run_id?, model_run_id?, config_hash?, actor_id	One primary origin owner appropriate to kind; parent references via edges
source.provenance_edges	project_id, child_provenance_id, parent_provenance_id, relation	Composite PK child,parent,relation; prohibit self-reference and cycles

Mapping HMAC keys are service secrets. Cross-release aliases may point to the same patient only with reviewed linkage. The identity group, not a synthetic branch ID, is the evaluation partition key. Age at baseline is an optional source-supported quantity in clinical observations; no invented birth date is needed.

6 Clinical records schema

6.1 Common observation envelope

clinical.observations contains id, project_id, patient_id, branch_id?, visit_id?, kind, concept_definition_id, time_reference_id, provenance_id, lifecycle, certainty, negation, supersedes_id?, source_locator_id?, recorded_at. kind is laboratory, vital, symptom, diagnosis, treatment, outcome, or note. certainty is documented, reported, uncertain, or unknown; negation is affirmed, negated, or unspecified. accepted observed source assertions require a source locator. An editor cannot label unsupported manual content REAL_OBSERVED.

clinical.observation_heads(project_id,patient_id,context_key,logical_id,current_observation_id,revision) is the mutable pointer to the current accepted revision. context_key is observed or the branch UUID text. Each revision chain has one head; serialize correction by locking this row. Historical snapshots link exact observation IDs, never a head pointer.

Table	Columns beyond common identity	Rules
clinical.visits	project_id, patient_id, branch_id?, label, time_reference_id, occasion_type, provenance_id	An occasion is not the timestamp of all its members
clinical.numeric_results	observation_id PK, value?, comparator?, threshold?, unit?, missing_reason?, original_value?, original_unit?, conversion_version?, reference_interval_id?	Only laboratory/vital kinds; value or explicit absence/censoring, not both
clinical.reference_intervals	project_id, concept_definition_id, low?, high?, unit, context jsonb, source_locator_id, version	At least one bound; low no greater than high
clinical.symptom_results	observation_id PK, presence, severity_value?, severity_scale_id?, onset_time_id?, resolution_time_id?	Absent symptoms cannot have positive severity; nonnumeric scales retain a code
clinical.diagnosis_results	observation_id PK, histology?, side?, grade?, clinical_stage?, pathologic_stage?, staging_system?, er_status, pr_status, her2_status	Receptors positive, negative, equivocal, unknown; unknown is not negative
clinical.treatment_events	observation_id PK, category, event_type, regimen_text?, setting?, started_time_id, ended_time_id?, exposure_status	Recorded versus hypothetical exposure distinct; no prescribing operation
clinical.outcome_results	observation_id PK, outcome_type, value_code?, value_text?, numeric_value?, unit?, assessment_method	pCR, imaging response and other outcomes have different codes
clinical.note_documents	observation_id PK, artifact_id, language, content_hash, note_type	Text bytes private; source span refers to exact content version

Table	Columns beyond common identity	Rules
clinical.retractions	project_id, patient_id, observation_id, reason_code, reason_text, actor_id	Append only; no physical mutation of source assertion
clinical.conflicts	project_id, patient_id, left_observation_id, right_observation_id, conflict_type, resolution_id?	Preserve both assertions and review decision

Each observation must have exactly one matching subtype; deferred constraint triggers enforce subtype/kind consistency. A clinical entry containing a panel creates one observation per analyte inside one atomic write. Uncertain facts extracted from a note live in extracted_facts until reviewed; extraction is not automatic promotion to accepted clinical truth.

7 Imaging and artifact schemas

ops.artifacts is the shared binary-object catalog: project_id, patient_id?, branch_id?, kind, media_type, object_key, object_version, byte_size, sha256, license_id, provenance_id, availability, created_by, recorded_at. availability is uploading, quarantined, verified, unavailable, or deleted. Object keys are private service data. Artifact lineage cannot weaken any parent source's rights or access constraints. Deduplicated bytes do not deduplicate authorization records.

Table	Columns	Scientific constraints
imaging.studies	project_id, patient_id, visit_id?, time_reference_id, source_mapping_id, protected_study_uid, modality, provenance_id	P0 modality MR; source UID uniqueness scoped to mapping/release
imaging.series	project_id, patient_id, study_id, protected_series_uid, sequence_label, provenance_id	Composite study ownership FK
imaging.acquisitions	project_id, patient_id, series_id, phase_index?, time_reference_id, volume_artifact_id, spatial_frame_id, channel_role	DCE phase remains inside study; not a follow-up visit
imaging.spatial_frames	project_id, patient_id, coordinate_system, spatial_unit, affine jsonb, shape jsonb, geometry_hash	coordinate LPS or RAS; unit mm; finite invertible 4 by 4 affine and three positive dimensions
imaging.lesions	project_id, patient_id, side, label, correspondence_status	Stable identity only when supported
imaging.lesion_correspondences	project_id, patient_id, prior_lesion_id, current_lesion_id, method, reviewer_id?, evidence_provenance_id	Explicit reviewed longitudinal linkage; split/merge relations retained
imaging.segmentations	project_id, patient_id, branch_id?, lesion_id, acquisition_id?, mask_artifact_id, spatial_frame_id, annotation_class, method, label_value, quality_status, provenance_id	Anatomical contour, functional analysis, approximate, bounding box; mask semantics explicit
imaging.tumor_measurements	project_id, patient_id, branch_id?, lesion_id, time_reference_id, segmentation_id?, acquisition_id?, source_locator_id?, metric_code, value, unit, method_version, comparability_key, quality_status, uncertainty_json?, provenance_id	At least one measurement source; volume and diameter use different codes
imaging.registrations	project_id, patient_id, fixed_frame_id, moving_frame_id, transform_artifact_id, method_version, quality_status, provenance_id	Transform direction and spatial units required
imaging.geometry_artifacts	project_id, patient_id, branch_id?, lesion_id, segmentation_id?, artifact_id, spatial_frame_id, display_transform jsonb, geometry_kind, source_measurement_id?, target_measurement_id?, actual_volume_mm3?, quality_status, provenance_id	geometry_kind observed_derived, synthetic, illustrative, interpolated, predicted

Table	Columns	Scientific constraints
imaging.qc_results	project_id, patient_id, target_artifact_id, check_code, check_version, outcome, details jsonb, reviewed_by?	Keep check-specific failures and warnings

The volume algorithm is voxel count times the absolute determinant of the affine spatial matrix, in mm^3. Mesh smoothing does not redefine source mask volume. A bounding annotation cannot be published as a measured anatomical contour. Store a target synthetic volume separately from the measured volume of the generated mask. Missing follow-up imaging has no manufactured study record. Illustrative surrounding anatomy must be labeled and cannot support patient measurements.

8 Twin and simulation schemas

Table	Columns	Constraints
twin.patient_states	project_id, patient_id, branch_id?, context, anchor_time_id, source_visit_id?, data_revision, snapshot_hash, assembly_version, publication_status, supersedes_state_id?, published_at?	context observed requires null branch; synthetic requires branch; snapshot content immutable
twin.state_observation_links	project_id, patient_id, state_id, observation_id, role, temporal_relation	Composite PK state,observation,role; role current or historical_context
twin.state_measurement_links	project_id, patient_id, state_id, measurement_id, temporal_relation	Exact immutable measurement version
twin.state_geometry_links	project_id, patient_id, state_id, geometry_id, display_role, temporal_relation	No silent last-scan carry-forward
twin.state_modalities	state_id, modality, availability, missing_reason?, quality, provenance_summary jsonb	PK state,modality; available, unavailable, invalid, pending, rejected
twin.scenario_definitions	project_id, code, version, rules_artifact_id, parameter_schema jsonb, config_hash, enabled	Unique project,code,version; SC-01 through SC-05
twin.simulation_branches	project_id, patient_id, baseline_state_id, scenario_definition_id, eligibility_mode, baseline_hash, status, published_run_id?	Baseline published and observed; frozen baseline hash
twin.simulation_runs	project_id, patient_id, branch_id, seed, parameters jsonb, config_hash, engine_version, container_digest, prng_version, environment_hash, status, output_hash?, job_id	Never overwrite an earlier execution
twin.branch_state_links	project_id, patient_id, branch_id, run_id, position, state_id, role	PK run,position; position zero is shared observed baseline; others own branch
twin.validation_results	project_id, patient_id?, target_resource_id, target_kind, rule_code, ruleset_version, severity, outcome, details jsonb	FK target to resource registry; fatal failure blocks publication
twin.publication_events	project_id, target_resource_id, action, actor_id, reason?, recorded_at	Append-only publication history

The branch baseline is linked, not copied into synthetic observations. Standard state member links never admit another branch's records. Publication checks explicitly allow the shared baseline at branch position zero. Branch eligibility_mode is constrained or unconstrained_research; P0 enables constrained only. Missing receptor evidence blocks an exposure rule that requires that evidence. Ruleset versions are reviewed project configuration, not clinical recommendations.

SC-01 encodes strong shrinkage; SC-02 modest change or stability; SC-03 growth; SC-04 shrinkage then regrowth; SC-05 clinical deterioration with independently specified tumor behavior. None automatically maps to a formal clinical response classification. Branch inputs may configure separate treatment events and symptom/laboratory dimensions.

Seed is a nonnegative signed-64-bit integer encoded as a JSON string. Reproducibility includes baseline hash, parameters, engine, PRNG, dependencies and numerical tolerance. Generated UUIDs, job times and database insertion order are excluded from scientific output hashes. Seed alone is not a reproducibility guarantee. Same science fingerprint may reuse a successful result after authorization; a requested fresh execution receives a new run ID.

9 Evidence inference and operations schemas

Table	Required columns and optional fields
intelligence.evidence_documents	project_id, title, canonical_url?, doi?, publication_date?, retrieved_at, document_version, content_hash, license_id, artifact_id, review_status
intelligence.evidence_passages	project_id, document_id, locator jsonb, text_hash, text_artifact_id
intelligence.model_registry	project_id, provider, configured_model_id, version_label, capability, input_schema_version, evaluation_artifact_id?, approved_export_policy_id?, enabled
intelligence.model_runs	project_id, patient_id, branch_id?, state_id, model_registry_id?, execution_mode, prompt_template_version, input_hash, corpus_version, status, usage jsonb?, job_id
intelligence.extracted_facts	project_id, patient_id, model_run_id, note_observation_id, concept_definition_id?, value jsonb, certainty, negation, time_reference_id?, span_start, span_end, span_hash, validation_status
intelligence.interpretations	project_id, patient_id, branch_id?, state_id, model_run_id, schema_version, content_hash, content jsonb, publication_status, generated_for_hash
intelligence.claims	project_id, interpretation_id, claim_kind, text, metric_value?, metric_unit?
intelligence.claim_record_links	project_id, claim_id, resource_id, relation
intelligence.claim_passage_links	project_id, claim_id, passage_id, relation
ops.resource_registry	project_id, id, resource_kind, patient_id?, branch_id?, provenance_id?
ops.processing_runs	project_id, patient_id, pipeline_code, code_version, parameters_hash, input_hash, job_id, status, output_hash?
ops.dependencies	project_id, input_resource_id, dependent_resource_id, input_hash
ops.jobs	project_id, patient_id?, branch_id?, kind, status, revision, progress?, result_resource_id?, error_code?, submitted_by, cancel_requested_at?, created_at, finished_at?
ops.job_attempts	job_id, attempt_number, lease_token, leased_until, worker_id, started_at, finished_at?, outcome?, error_code?
ops.outbox_events	project_id, aggregate_id, aggregate_revision, event_type, payload jsonb, delivered_at?
ops.idempotency_keys	project_id, actor_id, route, key, request_hash, response_status, response_json, expires_at
ops.audit_events	project_id, actor_id, action, target_id?, request_id, outcome, recorded_at
ops.upload_sessions	project_id, patient_id?, branch_id?, expected_sha256, expected_size, media_type, object_key, expires_at, status

Register each linkable resource atomically with its canonical row. Deferred triggers verify registry kind, ownership and row existence; a bare polymorphic UUID is not sufficient integrity. Dependency and provenance cycles are rejected. Embeddings are optional and must retain model identity, dimensions, corpus version and passage ID if introduced.

Text spans use zero-based Unicode code-point offsets, with end exclusive, over an exact normalized note version. Store the normalization version and content hash. JavaScript clients must convert UTF-16 offsets correctly. A source passage supports general background; it cannot establish an unrecorded patient fact. Claim validation checks existence, ownership, numerical agreement and meaningful support; a resolvable citation alone is insufficient.

10 SQL enforcement and query design

Use ordinary CHECK constraints for same-row rules, FKs for ownership, and deferred triggers/transactional services for cross-row checks. PostgreSQL CHECK constraints must not be used to pretend to validate other rows [R1]. The following is an enforcement pattern, not a complete migration:

```
CREATE TABLE twin.state_keys (
  project_id uuid NOT NULL,
  patient_id uuid NOT NULL,
  id uuid NOT NULL,
  branch_id uuid,
  context text NOT NULL,
  PRIMARY KEY (id),
  UNIQUE (project_id, patient_id, id),
  CHECK ((context = 'OBSERVED' AND branch_id IS NULL)
      OR (context = 'SYNTHETIC' AND branch_id IS NOT NULL)),
  FOREIGN KEY (project_id, patient_id)
    REFERENCES source.research_patients(project_id, id),
  FOREIGN KEY (project_id, patient_id, branch_id)
    REFERENCES twin.simulation_branches(project_id, patient_id, id)
);
```

Implement the same context guard on canonical patient_states; state_keys illustrates key enforcement and is not an additional mandatory table. Deferred branch/baseline checks handle creation ordering. Explicit non-null columns remain necessary because a SQL CHECK that evaluates to null does not reject a row.

Index source mapping uniqueness; patient_id and time references; branch_id and run position; project_id and publication_status; observations by patient, kind and recorded_at; state memberships by both sides; jobs by status and created_at; pending outbox by delivered_at; artifact object identity; claims by interpretation. Add a partial unique index preventing two accepted corrections from superseding one observation. Observation-head locking additionally protects simultaneous correction versus retraction.

Query event time through time_references with its basis and domain; do not sort mixed dates and relative days by text. Default timelines exclude drafts, rejected, withdrawn and unrequested synthetic contexts. Use stable ID tie breakers. PostgreSQL row-level security is recommended defense in depth, with a non-owner application role and transaction-local project settings reset on pooled connections. Service authorization remains mandatory; workers use restricted roles. No application role may UPDATE or DELETE accepted clinical content or published snapshot members.

11 Common HTTP contract

Base path is /api/v1/projects/{project_id}. TLS and authenticated user session or bearer token are required. Every nested ID is independently authorized and checked against its route owner. Return 404 for inaccessible patient resources to avoid existence disclosure; 401 for unauthenticated requests and 403 for authenticated users lacking an otherwise visible operation. Cookie sessions require CSRF protection and explicit origin checks for writes.

Use JSON with snake_case and OpenAPI 3.1.1 as a pinned contract version, not a claim that it is the latest version [R2]. Requests reject unknown fields. Nullable and omitted are distinct: an absent optional field leaves a draft field unchanged; null clears only a nullable draft field. Accepted scientific content is corrected through a new record. Responses may gain optional fields within v1; clients ignore unfamiliar response fields and display unknown future enum values safely.

UUIDs are strings; quantities are decimal strings; timestamps use RFC 3339 with offset; dates use YYYY-MM-DD; hashes are lowercase SHA-256. No NaN or infinity. Request bodies have a 1 MiB default limit; atomic observation bundles have at most 100 entries. Larger imports use private uploads and manifests. These are configurable prototype limits, not clinical limits.

Lists return items, next_cursor and snapshot_revision. limit defaults to 25 and has maximum 100. The opaque signed cursor binds filters, project, context, order and snapshot revision; expired cursors return 409 CURSOR_EXPIRED. Pages remain on that snapshot; later changes require restarting pagination. List filters include kind, publication_status, branch_id and time_domain where applicable. Unknown filters return 422.

Successful synchronous creation returns 201 and Location; completed command without a new resource returns 200; asynchronous work returns 202, Location to the job and Retry-After: 2. This is acceptance, not completion [R3]. Protected

data responses use Cache-Control: private, no-store. The session expires independently of a resource ID or cached response.

Mutating commands require Idempotency-Key, scoped to actor, project, method and canonical route. Retain keys for seven days, a project policy. Same key and canonical request hash replays the original status and body, including the same job ID. A different body returns 409 IDEMPOTENCY_KEY_REUSE. Concurrent identical requests serialize on a unique key; do not start duplicate work. Authorization is rechecked before replay. After expiry, the client must inspect the known resource before retrying a creation.

Mutable aggregate commands require If-Match using the strong ETag returned by GET, for example "patient-7" or "branch-3". Missing precondition returns 428; mismatch returns 412 [R3]. Version checks and increments happen under row lock in the committing transaction. Snapshot content ETags are immutable hashes; changing publication metadata changes the wrapper ETag. Idempotency replay takes precedence over stale If-Match when returning the already committed original operation.

12 Endpoint catalog

All paths below are relative to the project base unless stated. V, E, A mean viewer, editor and admin; higher roles include lower permissions. Patient-level restrictions still apply. GET detail resources return their corresponding canonical schema plus public metadata; internal columns such as object_key and encrypted mappings are never serialized.

Method and path	Role	Request or filters	Success
GET /api/v1/me	authenticated	none	200 identity and project roles
GET /patients	V	cursor, limit, source_id?, availability?	200 PatientSummary list
GET /patients/{p}	V	none	200 PatientDetail and ETag
GET /patients/{p}/visits	V	branch_id?	200 Visit list; observed default
GET /patients/{p}/observations	V	kind?, branch_id?, time_domain?	200 Observation list
POST /patients/{p}/observation-batches	E	ObservationBatch and If-Match	201 receipt with observation IDs and new revision
POST /patients/{p}/observations/{o}/corrections	E	replacement payload, reason, If-Match	201 new revision receipt
POST /patients/{p}/observations/{o}/retractions	E	reason, If-Match	201 retraction receipt
GET /patients/{p}/states	V	branch_id?, status?	200 StateSummary list
GET /patients/{p}/states/{s}	V	none	200 StateDetail
POST /patients/{p}/state-builds	E	StateBuildRequest, If-Match	202 JobAccepted
GET /patients/{p}/comparisons	V	from_state_id, to_state_id, metric_code, lesion_id	200 Comparison, including unavailable status
GET /patients/{p}/studies	V	none	200 ImagingStudy list
GET /patients/{p}/studies/{study}	V	none	200 study, series and acquisitions
GET /patients/{p}/segmentations/{seg}	V	none	200 Segmentation metadata
POST /patients/{p}/imaging-jobs	E	acquisition_id, segmentation_id, pipeline_version, If-Match	202 JobAccepted
GET /scenario-definitions	V	enabled?	200 versioned scenario list and parameter schemas
POST /patients/{p}/eligibility-checks	E	baseline_state_id, scenario_definition_id, exposure_config	200 EligibilityResult; no prescribing output
POST /patients/{p}/branches	E	BranchCreate, If-Match	201 BranchDetail in draft status

Method and path	Role	Request or filters	Success
GET /patients/{p}/branches/{b}	V	none	200 BranchDetail; only published for V
POST /patients/{p}/branches/{b}/runs	E	SimulationRunRequest, If-Match	202 JobAccepted
GET /patients/{p}/branches/{b}/runs/{r}	E	none	200 SimulationRun with validation links
POST /patients/{p}/branches/{b}/publications	E	run_id, validation_report_id, If-Match	201 publication event and branch status
POST /patients/{p}/interpretation-jobs	E	InterpretationRequest	202 JobAccepted
GET /patients/{p}/interpretations/{i}	V	none	200 Interpretation with freshness
GET /evidence/documents	V	query?, concept_code?, cursor?	200 approved evidence metadata list
GET /evidence/documents/{d}/passages	V	cursor?	200 approved passages within rights
GET /resources/{id}/provenance	V	depth 1–5	200 authorized nodes and edges; redact protected locators
GET /resources/{id}/validations	E	cursor?	200 rule results
GET /artifacts/{a}	V	none	200 safe ArtifactManifest
POST /artifacts/{a}/read-grants	V	purpose viewer or review	200 AssetGrant, audited and short lived
POST /upload-sessions	A	UploadRequest	201 UploadSession and signed upload instruction
POST /upload-sessions/{u}/completions	A	object_version, checksum	202 verification job
POST /imports	A	manifest_artifact_id, source_id	202 import validation job
GET /imports/{batch}	A	none	200 staged import and validation summary
POST /imports/{batch}/publications	A	validation_report_id, If-Match	201 publication receipt
GET /jobs/{j}	E	none	200 JobStatus; submitter/scope policy enforced
POST /jobs/{j}/cancellations	E	reason, If-Match	202 cancel-requested JobStatus
POST /resources/{id}/withdrawals	A	reason, If-Match	201 withdrawal event
GET /audit-events	A	target_id?, cursor?	200 redacted audit list

GET /health/live is unauthenticated and returns only status; GET /health/ready is restricted to the deployment monitor and checks essential dependencies. No public SQL, raw object proxy, free-form URL import, arbitrary code execution, or direct external-model endpoint is provided. P1 segmentation/prediction execution routes are absent until registered and evaluated; clients inspect capabilities instead of assuming availability.

13 Request and response structures

13.1 TimeReference and ObservationBatch

The TimeReference schema is a discriminated union on basis. timestamp requires value as date-time and time_domain; date requires value as date; relative_day requires decimal value and anchor_state_id; unknown requires missing_reason

and no value. anchor_state_id resolves to a persisted time reference after authorization. Synthetic relative times use the branch baseline anchor. Domain and branch ownership are checked, not inferred from the number.

POST observation-batches appends source-backed research observations. For P0, synthetic observations are worker outputs from simulation runs, not arbitrary editor writes. Branch editing is a future explicit capability. The client cannot set provenance, accepted status or data_revision; the service derives them from approved source artifacts and validated input.

```
{
  "schema_version": "observation-batch/1",
  "entries": [{
    "client_ref": "hb-1",
    "kind": "laboratory",
    "concept": {"system": "canceros-local", "code": "hemoglobin", "version": "1"},
    "time": {"basis": "date", "value": "2026-01-15", "time_domain": "source_shifted"},
    "result": {"value": "11.400000000", "unit": "g/dL"},
    "source_locator_id": "00000000-0000-4000-8000-000000000101"
  }]
}
```

```
{
  "batch_id": "00000000-0000-4000-8000-000000000102",
  "patient_id": "00000000-0000-4000-8000-000000000001",
  "data_revision": 8,
  "entries": [{"client_ref": "hb-1", "observation_id": "00000000-0000-4000-8000-000000000103"}],
  "state_update": "queued",
  "job_id": "00000000-0000-4000-8000-000000000104"
}
```

The request requires If-Match: "patient-7" and an Idempotency-Key. Response is 201 with ETag: "patient-8". All entries validate or none commit. The number is an example only; acceptance depends on configured analyte definitions and source verification. Diagnosis, symptom, vital, treatment, outcome and note variants use the subtype fields in section 6, a kind discriminator, time and source locator. Note input references an already verified private text artifact. Unsupported units return 422; abnormal but valid source results remain preserved.

Correction body is {reason_code, reason_text, replacement}; replacement uses the matching observation variant without an observation ID or ownership fields. It must reference source evidence and may change clinical time. Retraction body is {reason_code, reason_text}; both require patient If-Match. Changing kind requires a new observation and retraction rather than silently changing a laboratory result into a treatment event.

13.2 State build and snapshot

StateBuildRequest requires anchor_time, expected_data_revision and context=OBSERVED. Optional visit_id must belong to that patient and time context. No user-selected synthetic context is accepted through this route. Builder uses snapshot isolation to resolve exactly the requested data revision; if no longer buildable it returns 409 REVISION_UNAVAILABLE. A source revision may be old but must be explicit. Publication verifies the computed input hash and membership.

```
{
  "state_id": "00000000-0000-4000-8000-000000000020",
  "patient_id": "00000000-0000-4000-8000-000000000001",
  "branch_id": null,
  "context": "OBSERVED",
  "anchor_time": {"basis": "relative_day", "value": "0", "time_domain": "source_relative",
"anchor_state_id": "00000000-0000-4000-8000-000000000010"},
  "data_revision": 8,
  "publication_status": "published",
  "tumor_measurements": [{"measurement_id": "00000000-0000-4000-8000-000000000021", "metric_code":
"lesion_volume", "value": "12400", "unit": "mm3", "provenance_type": "REAL_DERIVED", "quality":
"pass"}],
  "geometry": [],
  "modalities": [{"modality": "geometry", "availability": "pending", "quality": "pending"}, {"modality":
 "laboratory", "availability": "unavailable", "missing_reason": "not_recorded", "quality":
"not_applicable"}],
  "snapshot_hash": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa",
  "warnings": ["GEOMETRY_NOT_READY"]
}
```

The example assumes a pre-existing anchor state; a baseline uses its own persisted source time reference, never a self-referencing anchor. Actual StateDetail additionally returns observations, treatment_context, provenance links and temporal_relation for each measurement and geometry link. Empty lists mean no eligible members, with availability explaining why. Historical geometry can be shown only with its actual time, relation=historical_context, and an explicit display notice. A lab-only update does not assert a current tumor scan.

13.3 Branch and simulation requests

BranchCreate requires baseline_state_id, scenario_definition_id, eligibility_mode=constrained and exposure_config. exposure_config contains category, setting_code and events with relative timing; it cannot include dosage recommendations. EligibilityResult returns eligible, ineligible or insufficient_data; ruleset_version, missing_fields, reasons and baseline_hash are required. A draft branch may exist while insufficient; runs return 409 until resolved through source-supported data.

```
{
  "seed": "42",
  "engine_version": "slre/1.0",
  "parameters": {
    "schema_version": "slre-parameters/1",
    "followup_days": [30, 60, 90],
    "tumor_pattern": "sustained_shrinkage",
    "clinical_condition": "stable"
  },
  "expected_baseline_hash": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa"
}
```

Scenario parameter schema defines bounds, required variables, unit dictionary, noise, missingness and geometry tolerance. Unknown parameter keys fail. Rules must be configured before this example is executable. The timeline includes baseline day zero plus three generated states. New runs never replace existing published run content; republishing changes the branch's current-run pointer with an audit event. Readers may retrieve a prior run by ID under permission.

13.4 Jobs and artifact access

JobAccepted is {job_id, status, status_url, result_url:null}. JobStatus adds kind, attempt_count, progress:null-or-number, result_resource_id:null-or-UUID, error:null-or-{code,retryable}, cancel_requested, created_at and finished_at. progress is monotonic within an attempt, not a fabricated percentage when work is unmeasurable.

ArtifactManifest includes artifact_id, kind, media_type, byte_size, sha256, availability, provenance_type, quality, spatial_frame? and rights_summary. AssetGrant is {artifact_id,url,expires_at,method:"GET"}. Default expiry is 120 seconds, configurable; never persist signed URLs as artifact identities or include them in AI prompts or logs. Authorization and license policy run on every grant. A signed URL may remain valid until expiry after a withdrawal; immediate revocation requires an authenticated proxy or provider revocation feature. Viewer role does not automatically authorize original DICOM downloads.

UploadRequest requires patient_id when patient-owned, kind, media_type, byte_size, sha256 and source_id. The service chooses a random quarantine object key; the client supplies no bucket path. Limit 512 MiB per object for P0; split larger datasets by manifest or use an admin batch importer. UploadSession returns upload_session_id, expires_at and fixed method/headers. Completion verifies the actual object version, byte count, hash and format before any import can publish it. Treat filenames, archives, DICOM metadata and note text as untrusted input.

13.5 Comparison contract

Comparison returns status comparable, not_comparable, or insufficient_data; from/to state IDs, lesion ID, metric_code, units, source measurement IDs, absolute_change?, percentage_change?, reason_codes and context. Comparable requires matched lesion meaning, metric, units, mask definition and supported time domain. Percentage is null with ZERO_BASELINE when baseline is zero. Do not divide volume by diameter or compare a functional analysis region against an anatomical volume. Mixed observed/synthetic comparisons explicitly return comparison_context=hypothetical_vs_observed. Comparisons are descriptive, not estimates of treatment efficacy.

13.6 Interpretation contract

InterpretationRequest requires state_id, input_snapshot_hash, purpose=research_summary and use_external_model boolean; optional comparison_state_id. External requests require approved export policy and model configuration. Policy refusal returns 403 EXTERNAL_PROCESSING_NOT_ALLOWED; provider failure can produce a deterministic summary with execution_mode=deterministic_fallback, clearly identified. No unavailable model result is forged.

Interpretation responses contain interpretation_id, state_id, generated_for_hash, freshness, execution_mode, model_identity?, corpus_version, findings, limitations and evidence. Each finding contains kind, text, observation_refs, measurement_refs and passage_refs, all authorized and persisted. Numeric findings also include metric_code, value and

unit taken from deterministic calculations. Unknown reference IDs or unsupported patient-specific claims block publication with FAILED_GROUNDING. Model output remains private until validated. Empty findings with status=insufficient_evidence is valid.

Freshness is current_for_snapshot when the hash matches the selected snapshot. It can simultaneously be superseded_for_current_timeline if later data changed that timeline. Historical interpretations remain valid records of their exact historical inputs; they are not silently rewritten. Forecast outputs and interpolated states are reserved P1 contracts, disabled in P0, and never placed in observed state members.

14 Errors and validation feedback

Errors use application/problem+json with RFC 9457 fields and bounded extensions [R4]. type is a stable URI under the deployed API's /problems namespace; do not expose stack traces, database names, patient source keys or private object paths.

```
{
  "type": "/problems/observation-validation",
  "title": "Observation validation failed",
  "status": 422,
  "detail": "One entry uses an unsupported unit.",
  "instance": "/api/v1/requests/req-example",
  "code": "UNSUPPORTED_UNIT",
  "request_id": "req-example",
  "errors": [{"pointer": "/entries/0/result/unit", "code": "UNSUPPORTED_UNIT"}]
}
```

The relative problem URI resolves against the deployed API origin; deployed named problems use their documented type URI and title. Stable code, not human detail, drives client behavior.

Status	Codes and handling
400	MALFORMED_JSON, INVALID_CURSOR; repair syntax
401 / 403 / 404	AUTHENTICATION_REQUIRED, OPERATION_FORBIDDEN, RESOURCE_NOT_FOUND; no resource leakage
409	IDEMPOTENCY_KEY_REUSE, INVALID_TRANSITION, BASELINE_CHANGED, INPUT_STALE, INELIGIBLE_SCENARIO; inspect current resource
412 / 428	PRECONDITION_FAILED, PRECONDITION_REQUIRED; refresh aggregate and retry deliberately
413 / 415	PAYLOAD_TOO_LARGE, UNSUPPORTED_MEDIA_TYPE
422	INVALID_TIME, UNIT_MISMATCH, CROSS_CONTEXT_REFERENCE, FAILED_SCHEMA; field pointers returned
429	RATE_LIMITED with Retry-After; do not create duplicate jobs
503	DEPENDENCY_UNAVAILABLE; retry boundedly using same idempotency key

Failures after 202 are recorded on the job, not changed into an unrelated HTTP success with invented results. A comparison that cannot be made is a 200 domain result with null calculations and explicit reasons; malformed comparison input is a 422 error.

15 State transitions and publication

Resource	Allowed transition	Guard and atomic effect
Artifact	uploading → quarantined → verified	Actual bytes and metadata checks; remain quarantined on failure
Import	staged → validating → ready → published	Approved source, rights, identity and required QC; publication commits rows and outbox
Import	validating → rejected	Record rule failures; no viewer-visible partial patient

Resource	Allowed transition	Guard and atomic effect
Observation	draft → accepted or rejected	Source and type validation; accepted content immutable
State	draft → validating → ready → published	Complete selected membership, hash and required checks
Branch/run	draft → queued → running → validating → ready	Frozen inputs, eligibility and all required checks
Branch	ready → published	User publication command; all required generated positions present
Interpretation	queued → running → validating → published	Input policy and claim grounding pass
Published resource	published → withdrawn	Authorized reason, audit, no silent replacement
Job	queued → running → succeeded or failed	Leased attempt; success requires durable result commit
Job	queued/running → cancelled	Cancellation acknowledged before publication commit

Job cancel_requested is a flag while running; it is not an immediate terminal result. Retry transient errors inside a job with a new attempt and lease; failed terminal jobs require a new authorized command/job. Maximum three attempts, exponential backoff with jitter, and configurable deadline are initial project policy. Validation failures are not retried blindly.

Late worker results must present the current lease token. Commit result rows, publication pointer and job success in one transaction only if the lease and cancellation guard still hold. Unique job-output constraints prevent duplicate visible output after redelivery. Queue delivery is at least once; this design makes publication idempotent rather than claiming exactly-once execution.

16 Incremental updates and transactional behavior

16.1 New clinical observation

1. Authenticate, authorize patient, validate idempotency and If-Match, and lock the patient aggregate.
2. Resolve concept, source locator, time and ownership; validate the whole bundle.
3. Insert immutable observations, subtype rows, provenance, heads and registry entries; increment data_revision.
4. Insert audit and outbox events in the same transaction. Commit once; return the receipt.
5. Outbox dispatcher queues only affected projections. A laboratory result rebuilds clinical summaries and related interpretations; it does not regenerate an unchanged lesion mesh.
6. Build a new immutable state for the selected revision. UI keeps the prior published state until the replacement is ready, shows processing status and never implies old geometry was collected at the new lab time.

The outbox closes the database/queue failure gap. Redis is transport, not the authoritative job or clinical store. Dispatcher redelivery is deduplicated by event and work fingerprint. Failed dispatch is recoverable by scanning pending outbox rows.

16.2 Late data and correction

Event time determines position in a compatible clinical chronology; recorded_at determines when CancerOS learned the fact. A late result may cause new snapshots and comparisons at affected anchors; it never changes the bytes or memberships of existing snapshots. Corrections append a replacement and update the current observation head under lock. A retraction changes the current projection through a separate event. Return 409 CORRECTION_CONFLICT if the target is already superseded or retracted.

Dependency traversal marks affected latest-timeline results as needing recomputation. It does not invalidate an unchanged historical snapshot's internal hash. A branch remains frozen to its original baseline; display baseline_superseded=true if later source evidence changed. Regenerating from a corrected baseline creates a new branch. No automatic rebasing or cross-branch mutation occurs.

16.3 Geometry update

A new mask or registration creates new artifacts and measurements. The worker writes to a unique immutable object version, verifies it, inserts catalog/results, and publishes only after QC. Object upload and SQL cannot share one atomic transaction: a failed SQL commit leaves a private orphan for reconciliation. A committed manifest with missing bytes is marked unavailable and blocks new read grants. Sweeper deletes only proven unreferenced objects older than the configured grace period.

16.4 Simulation and interpretation race

A run pins a baseline snapshot hash, not a mutable patient latest pointer. A successful run publishes the complete follow-up set atomically; partial follow-up rows stay invisible to viewers. Interpretation input pins state and evidence versions. If the requested snapshot hash mismatches, return 409 before execution. A later clinical update does not mutate that execution's inputs; its result is displayed against the pinned snapshot with current-timeline freshness metadata.

Surgery is a discrete event. An absent resected lesion is represented as absent/not applicable with an intervention context, not an empty mask mislabeled failed segmentation or a smoothly shrinking unresected tumor. Residual measurements require a documented definition. Such paths are enabled only when their scenario rules and representation are implemented.

17 Access security and inference boundaries

Viewer reads published approved summaries and eligible asset grants. Editor adds source-supported observations, runs approved simulations and requests interpretation. Administrator approves imports, configures sources/models, reviews rights and withdraws resources. Worker credentials authorize narrow job kinds and result publication, not unrestricted user administration.

The external inference gateway receives only policy-approved de-identified structured snapshots and approved evidence passages. It receives no database credentials, object signing credentials, arbitrary SQL tools or unrestricted note archive. Record export policy ID, input hash, model identity, prompt template version and execution status. Do not log raw notes, image tags, protected source mappings, signed URLs or provider secrets. Render note and model text as untrusted content; it cannot instruct services to bypass authorization or execute tools.

Import publication requires a rights review for the intended use, not just a non-null license_id. Processed descendants preserve parent restrictions and attribution requirements. Technical quality does not grant redistribution permission. Access and retention policies require project-specific review before any real sensitive data is used.

18 Migration and implementation deliverables

Implement Alembic migrations in this order: access and source identity; artifacts and registry; clinical observations/time; imaging; twin/simulation; evidence/inference; jobs/outbox and audit; enforcement triggers and indexes. Cyclic baseline/branch relations use deferred constraints with post-insert checks. Never introduce unvalidated cross-row assumptions through a permissive JSON field.

Use an expand-and-contract approach for populated schemas: add compatible columns and serializers, backfill with audit, enforce constraints after validation, then retire old fields in a later release. Keep rollback compatible with running services. Backups must include relational data and referenced object versions; perform a restoration test before the demo.

The implementation repository shall contain numbered migrations, Pydantic request/response models, generated OpenAPI 3.1.1, generated TypeScript clients, versioned scenario parameter schemas, minimal synthetic fixtures, authorization and integrity tests, and a migration/readme runbook. This specification is the contract baseline; it is not a claim that those executable artifacts already exist. P0 can omit optional embeddings, forecasting, interactive mask edits and specialized graph storage.

19 Contract acceptance tests

Test	Required outcome
Cross-patient or cross-project study/lesion/state reference	API rejects and database FK/trigger rejects direct invalid insert
Branch A observation linked to branch B or observed state	Rejected; shared baseline accepted only through designated baseline link
Missing stage, receptor, laboratory or source date	Remains unknown/unavailable; no invented defaults
DCE phase import	Adds acquisition inside existing examination; no fake follow-up visit
Same idempotency key, same request, concurrent submission	One durable mutation/job, same response replay
Same key with different body	409; no mutation
Concurrent writers using one patient ETag	One commits; other receives 412 with no partial writes
Failure inside observation bundle	Entire bundle rolls back, including revision, audit and outbox
Queue unavailable after observation commit	Durable outbox later dispatches; record is not lost
Worker dies or lease expires after output upload	Retry cannot double-publish; orphan reconciled privately
Cancellation races with worker publication	One guarded terminal outcome; cancelled work has no published partial output
Correction or late observation	Original record and snapshots unchanged; new current projection and freshness flags
Lab-only addition	No mesh rerun; prior geometry time remains visible
Unknown or shifted time compared to incompatible domain	Explicit not_comparable or rejection; no invented interval
Zero comparison baseline or incompatible measurement meanings	Percentage null with reason; no invalid number
Bounding annotation or failed mask QC	No published anatomical mask-derived volume claim
Repeated scenario with same complete science fingerprint	Same normalized scientific output within declared tolerance
Five scenario families	Each creates baseline plus at least three valid follow-ups with separate branch identity
Unavailable source fields in simulation	Synthetic values remain labeled; baseline source remains unchanged
Missing/false evidence or fabricated numeric AI claim	Interpretation not published; visible grounding failure
External export denied or provider unavailable	No unauthorized payload leaves; source views still work
Unauthorized original image grant	Denied despite knowledge of artifact UUID
Source correction after branch generation	Old branch remains frozen and is flagged baseline_superseded
Restore database and objects	Published manifests resolve to the correct verified object versions

Scientific checks additionally verify physical affine scaling, orientation, mask semantics and longitudinal lesion correspondence on curated fixtures. Passing API tests alone does not establish a valid real longitudinal demonstration. Release evidence must identify the actual matched case, source rights, usable dates, annotation definitions and measurement compatibility.

20 Traceability and references

Input	Contract coverage
COS-PRD-001 Product Requirements	Patient journeys, observation updates, role boundaries, four-position branch timeline and acceptance gates
COS-SPEC-003 Clinical Data and Dataset	Source identity, licenses, clinical subtypes, units, time precision and provenance
COS-SPEC-004 Synthetic Longitudinal Data	Five scenario families, baseline lineage, seeds, eligibility, validation and branch publication
COS-SPEC-005 Medical Imaging and Digital Twin	Study/acquisition distinction, spatial frames, masks, measurements, geometry and incremental updates
COS-SPEC-006 AI and Multimodal Intelligence	Missing modalities, note spans, evidence, model versioning, claim grounding and abstention
COS-SPEC-007 System Architecture	Modular FastAPI boundary, PostgreSQL, private artifacts, asynchronous workers and provider adapters

Technical references checked 22 October 2026. These support standard mechanisms, not CancerOS clinical claims or project-specific policy values.

- R1 PostgreSQL 18 Constraints: https://www.postgresql.org/docs/18/ddl-constraints.html
- R2 OpenAPI Specification 3.1.1: https://spec.openapis.org/oas/v3.1.1.html
- R3 HTTP Semantics RFC 9110: https://www.rfc-editor.org/rfc/rfc9110.html
- R4 Problem Details for HTTP APIs RFC 9457: https://www.rfc-editor.org/rfc/rfc9457.html

Open implementation decisions are the verified source case inventory, mask annotation classes, approved source rights, reviewed scenario eligibility rules, enabled inference provider and export policy, and measured worker/upload limits. None may be filled by fabricated patient facts, assumed model availability, or invented clinical thresholds.